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Living on the edge: network neuroscience beyond nodes

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Abstract

Network neuroscience has emphasized the connectional properties of neural elements – cells, populations, and regions. This has come at the expense of the anatomical and functional connections that link these elements to one another. A new perspective – namely one that emphasizes 'edges' – may prove fruitful in addressing outstanding questions in network neuroscience. We highlight one recently proposed 'edge-centric' method and review its current applications, merits, and limitations. We also seek to establish conceptual and mathematical links between this method and previously proposed approaches in the network science and neuroimaging literature. We conclude by presenting several avenues for future work to extend and refine existing edge-centric analysis.

Network neuroscience beyond nodes

Considerable effort has been expended in mapping brain connectivity across spatiotemporal scales [1–4]. Connectivity patterns can be represented as a network composed of nodes and edges corresponding to neural elements – cells, populations, and areas – and their functional and anatomical connections, respectively [5].

Most analyses of brain networks have focused on nodes, and have quantified their degree, centrality, or communities. This parallels a rich history of 'brain mapping' wherein neural elements are annotated with their cytoarchitectural, functional, and neurochemical properties [6,7]. There are many reasons why this 'node-centric' emphasis may be warranted. First, it mirrors at least a century of empirical observations demonstrating that nodes constitute the fundamental unit of processing/computation, whereas edges relay the results

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Declaration of interests

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of computations. Second, it offers a natural way to decompose the brain into functionally specialized units, thereby facilitating fundamental discoveries concerning brain organization [8–11] while reifying well-established neuroscientific concepts [12].

Nonetheless, two decades of sustained effort based on node-centric models have left numerous questions unanswered. How does system-level architecture emerge from time-varying fluctuations in activity/connectivity? To what extent are systems overlapping or distinct? What are the origins of brain 'fingerprints'? How might they be capitalized upon to improve brain-based phenotyping?

Efforts to address these questions may benefit from a shift in perspective. One possibility is to consider **higher-order network structure** (see Glossary) – where structure is quantified by using hypergraphs, simplicial complexes, or multivariate information theory. Alternatively, we may shift focus onto modeling frameworks that are compatible with existing data structures and retain familiar and neurobiologically informed conventions/definitions such as **structural connectivity** and **functional connectivity (FC)**, while extending the concept of connectivity to higher dimensions. This framework can be considered to be 'edge-centric' in that it characterizes interactions between the edges of a network rather than its nodes (Box 1). It yields two useful constructs: (i) edge time-series as an instantaneous estimate of **edge weight**, and (ii) edge FC as the similarity matrix between all pairs of edges. In this review we cover the procedure for constructing both aspects, review the growing list of findings made using edge-centric networks, and discuss their future applications and extensions as well as potential limitations.

Edge-centric networks

Constructing 'classical' edge-centric networks

The network science literature is replete with approaches for constructing edge-centric networks. The two best-known are the so-called **link similarity** and **line graph** approaches (Figure 1A–C) [13,14].

Both transform a $N_{nodes} \times N_{nodes}$ connectivity matrix, W_N , into a $N_{edges} \times N_{edges}$ matrix, W_E , whose elements encode the strength of association between the edges of W_N . In the case of link similarity, edges that share 'stubs' – the two nodes linked by a given edge – are associated with one another based on the normalized overlap of their 'unmatched stubs' connectivity profiles (Figure 1B). This procedure is repeated for all pairs of edges, resulting in a 'link similarity matrix' whose rows and columns correspond to edges in the original network. Line graphs are constructed by representing W_N as an incidence matrix, H (Figure 1C). Rows in H correspond to edges and contain entries of zero except for the columns corresponding to nodes joined by a given edge. The product HH^T generates a 'line graph', W_E , where edges are connected to one another if they have a shared stub.

The matrices generated using these methods are edge-centric in the sense that the units of interest – the nodes in W_E – correspond to edges in W_N . Edge-centric networks can be treated identically to node-centric networks, although they possess some unique advantages. Local measures in W_E are mapped to edges in W_N . This has been especially useful in

community detection – the algorithmic partitioning of networks into subnetworks [15–18]. When applied to W_E , each edge is assigned to a non-overlapping community. However, edge communities can be projected onto nodes, allowing for overlapping nodal communities (Figure 1D).

Edge-centric networks for time-series data

Although edge-centric methods have existed for over a decade, applications to brain network data have been limited [19,20]. Existing methods are memory-intensive and operate on sparse networks with positive edge weights, thus exposing a tension with network neuroscience where FC is typically defined as fully weighted and signed.

How might we generate an edge-centric network that captures the spirit of line graphs/link similarity but is also self-consistent with established methods for defining node-centric brain networks? One approach is to temporally 'unwrap' FC (Box 2). This procedure results in a co-fluctuation time-series $r_{xy} = [x_z(1) * y_z(1), \dots, x_z(T) * y_z(T)]$ for every pair of nodes $\{X, Y\}$. The variable $X_{z(t)}$ represents the value of the z -scored time-series for node X at time t .

$r_{xy(t)}$ carries temporally resolved information about the weight of edge $\{X, Y\}$ and can be interpreted as a measure of **time-varying functional connectivity (tvFC)**. Sliding-window connectivity is the most popular method for estimating tvFC. Briefly, and in its most generic form, this method centers a window of length W on frame t , such that the $\frac{W}{2}$ preceding and following frames all fall within the window. Then, using only those observations, FC is estimated between all pairs of nodes. The window is advanced a fixed number of frames (often by only a single frame, such that adjacent windows have $W - 1$ frames in common), and the procedure repeated. Doing so for all timepoints yields a series of temporally resolved connectivity matrices [21–23]. Note that this procedure requires the user to specify parameter values for the window length, the shape of the window (e.g., boxcar, Gaussian, exponential), and the amount of overlap between successive windows. Unlike these window-based approaches, estimating **edge time-series (eTS)** is parameter-free and requires no windowing.

One can estimate eTS for every unique edge, resulting in a matrix with dimensions $N_{edges} \times N_{frames}$. Using this set of time-series, we can derive an edge-centric network based on the similarity of eTSs to one another (Box 2). Given its structural similarity to FC, we refer to this construct as **edge functional connectivity (eFC)** [24]. The eFC matrix is much larger than its node-centric counterpart, with dimensions of $N_{edges} \times N_{edges}$, and is fully weighted and signed.

The first study to investigate eFC analyzed fMRI data from three cohorts [24]. The authors found that, like classic FC [25], eFC matrices stabilized given sufficient samples and exhibited high levels of subject specificity, thus enhancing identifiability beyond the levels obtained using nodal FC [26]. The authors also reported that the eFC was state-dependent because it dissociated resting-state from movie-watching data.

The most interesting feature of eFC is its **community structure** [24,27,28], which refers to the algorithmic partitioning of the elements of a network into meaningful subnetworks [29]. When projected onto nodes, edge communities result in overlapping communities. Although this property is not uncommon across community detection algorithms [30–33], it remains rare in network neuroscience where modularity maximization and Infomap are field standards [15,16]. Those approaches continue to be used for delineating boundaries between 'canonical' brain systems [9,34,35]. However, when combined with a functionally evocative naming convention – for example, labels such as 'visual community' or 'somatomotor module' – this can falsely give the impression that the boundaries of a community completely circumscribe all territories associated with its named function and, conversely, that the community itself performs only a single or narrow range of functions [36]. Moreover, the reliance on hard partitions tends to overstate the degree of functional specialization and understate the need to reconfigure or break modular boundaries in the service of integrated brain function [37–39]. Network models that capture the overlap between competing sets of communities can naturally capture this important aspect of functional organization.

What are the topographic profiles of edge communities? Broadly, they resemble those reported using classical techniques [8,9,34]. However, the extent to which nodes participate in edge communities is graded, analogously to spatial independent component analysis (ICA) maps [40]. That is, edge communities 'draw in' other collections of nodes to well-known large-scale networks, forming new integrated constructs (Figure 2). A particularly salient example is the default mode network edge community whose topography closely resembles that of the canonical network, but only weakly incorporates nodes from sensorimotor systems [24]. In this way, we can think of the edge communities as a refinement of traditional node-level communities. Nodal communities are an excellent zero-order approximation that captures the strongest and most salient aspects of the community structure of the brain. However, they miss out on fine-scale structure, including partial community affiliations, which can be captured by edge communities which allow for overlap.

The extent of community overlap varies across the brain and can be quantified using a normalized entropy measure [24]. Surprisingly, the highest levels of overlap were concentrated within sensorimotor and attentional systems, and association networks exhibited reduced entropy. This pattern contrasts with the extant literature where measures such as participation coefficient and other measures of community overlap have implicated association cortices as having the greatest level of overlap [41,42].

eTS as an estimate of tvFC

Estimating eFC creates 'edge time-series' as an intermediate construct. One of the most useful properties of eTS is as an estimator of tvFC [43,44]. The earliest tvFC studies used sliding windows to obtain 'dynamic' estimates of interregional correlation/coherence [38,45,46]. Although popular, this approach has drawbacks, including poor temporal localizability and the need to specify multiple parameters.

eTS helps to address localizability concerns. Calculating eTS for all edges results in a $N_{edges} \times N_{frames}$ matrix (Figure 3A, top). The columns of this matrix index confluctuation values for all edges at a given time. These values can be rearranged to form the upper triangle of a $N_{nodes} \times N_{nodes}$ matrix (Figure 3A, bottom), yielding a temporally resolved estimate of whole-brain confluctuations. Unlike sliding-window methods, eTS are estimated without parameterization or windows. Indeed, eTS is only modestly associated with sliding-window tvFC [47]. We note that this correspondence improves up to a point as window size decreases and sliding-window estimates approach the single-frame timescale of eTS. However, for very small window sizes, the correlation estimates obtained from the sliding-window technique become highly uncertain and the correspondence decreases.

eTS approaches have other useful properties. By construction, the temporal average of eTS is exactly the product–moment correlation. At the whole-brain level, this procedure does not generate only one coefficient, and instead generates a vector of coefficients ($1 \times N_{edges}$) that can be rearranged into a $N_{nodes} \times N_{nodes}$ matrix. This matrix is exactly static FC; eTS are a precise temporal decomposition of FC. Note that, in principle, eTS can be smoothed using a window/kernel, although this procedure likely blurs fine-scale temporal details, and the smoothed estimates are no longer an exact decomposition of static FC.

Another feature of eTS is their apparent 'burstiness'. In line with the extant literature [48–51], long periods of relative quiescence are punctuated by brief intervals of high-amplitude confluctuations that are not associated with movement or physiological confounds [52,53]. These bursts appear as vertical striations in the eTS matrix and correspond to brain-wide **events** and peaks in the root mean square (RMS) time-series (Figure 3A, middle). These extreme confluctuations are largely absent in randomized (circularly shifted) data [53,54].

We can gain insight into the importance of events by 'filtering' frames to obtain estimates of high/low-amplitude FC. High-amplitude FC is more modular and correlates better with static FC, suggesting that events explain the system-level organization and coupling structure of the brain (Figure 3B). Events can also be clustered into **brain network states** (Figure 3C) [53,55,56]. The largest cluster is characterized by confluctuations of default and salience/ventral attention networks. When projected onto nodes, this confluctuation pattern is immediately recognizable as the default mode network [57], first principal gradient [58], sensorimotor–association axis [59], or extrinsic/intrinsic or task-positive/negative division [60,61]. The second cluster is characterized by opposed confluctuations of control and dorsal attention networks. These states typically account for ~50% of events and have been observed in multiple datasets [3,25,62,63]. There is less consistency surrounding the cluster structure of remaining events, suggesting that they may be kinless, with no clear cluster affiliation.

What factors are necessary for events to emerge? One approach for addressing this question is through *in silico* simulations. A recent study coupled oscillators to one another based on empirical structural connectivity [54]. When simulated, this system generates phase time-series for each node. These phases were hemodynamically convolved to generate blood oxygen level-dependent (BOLD)-like activations that exhibited events. Interestingly, confluctuation patterns expressed during events aligned with anatomical modules. Event

counts dropped when that modular structure was degraded, suggesting that anatomical modularity may be crucial for organizing brain activity into cohesive cofluctuations. Another study implemented a more detailed biophysical model, and established a link between events and the ignition/propagation of cascades in population activity [64].

These modeling studies imply that mechanisms driven by variations in internal state or external stimulation are not necessary for creating momentary fluctuations in eTS, including events. Such fluctuations can emerge spontaneously from the underlying dynamics. They also rule out physiological confounds and motion as singular explanations.

On the other hand, several studies have explored the origins of events empirically using naturalistic stimuli [65,66]. One such study found that RMS time-series exhibited strong inter-subject correlations during movie-watching [52], whereas another showed that scene changes consistently elicited events [67]. These studies suggest that event timing may be modulated by sensory input [68,69].

In short, there is evidence that events can not only occur spontaneously but also in response to stimuli. Consequently, a mechanistic description of the origins of events remains to be elucidated. A possible hint comes from task studies. During cognitively demanding tasks, activations 'quench' FC, reducing variability in activity while weakening the magnitude of FC [70]. One recent study observed that, during movie-watching, events reliably occurred near the ending of movie scenes, mimicking a transition to a rest-like state that could be accompanied by increased cofluctuation amplitude [67].

eTS for studying individual differences

One of the aims of neuroscience is to understand how variation in activations, morphology, and connectivity translate into differences in behavioral, cognitive, and demographic profiles [71]. In network neuroscience this often means identifying elements of brain networks that covary with behavior. Some of these features are state-dependent – for example, task-evoked connectivity. Other features are more stable, trait-like, and unique to an individual. Indeed, many studies have focused on network-based 'fingerprinting' – the enterprise associated with identifying network features based on the extent to which they are individualized [72,73]. This often entails estimating a set of brain-based features (e.g., connection weights) from a series of scans in which individuals are represented more than once. That is, each participant is scanned at least twice. The expectation is that those features render participants 'identifiable' – in other words, they are more similar to other scans of themselves than to scans of other participants [74].

eTS may be useful for studying individual differences. For instance, one of the earliest findings was that, compared to low-amplitude frames, high-amplitude frames appeared to carry more subject-specific information. That is, based on the cofluctuation patterns expressed during high-amplitude frames, subjects can be reliably distinguished from one another [52,53]. Indeed, it might even be possible to enhance the fingerprints by reconstructing FC from a select set of frames [75] not limited to high amplitudes. By contrast, the low-amplitude frames carry relatively little information about an individual and are more likely to be associated with in-scanner motion [53].

A growing number of studies have used eTS to study individual differences. The first eTS study analyzed Human Connectome Project (HCP) data, derived factor scores from behavioral/demographic data, and then calculated correlations with FC reconstructed from high/low-amplitude frames [52]. They found that the high-amplitude frames exhibited stronger correlations on average. Although recent studies have shown that, for select sets of behavioral measures, middle-amplitude frames (non-events) were most useful for predicting behavior [76,77].

Other studies have used eTS to characterize brain recovery from stroke [78], classify individuals with autism spectrum disorder [47,79], investigate episodic memory [80], study communication profiles of basal ganglia with cortex [81], study time-varying structure-function coupling [82], link event clusters with endogenous hormone fluctuations [56], and distinguish between task states [83].

In summary, eTS are a parameter-free method for decomposing FC into time-varying contributions. Frames in the eTS matrix are networks whose weights correspond to instantaneous cofluctuation magnitude. Because mean eTS resolve to static FC, we can identify frames that best explain FC. High-amplitude events stand out from the others. Evidence suggests that, although events account for a small fraction of frames, they carry subject-specific information. We note that events are not the complete story. Several studies have suggested that middle- and low-amplitude frames may be more predictive of cognitive ability [76,77] than FC reconstructed from events. One possible explanation is that events promote a pattern of cofluctuation that, although idiosyncratic, is shared across individuals. By excluding events, one can amplify inter-individual variation and enhance brain-behavior associations. Future work will be necessary to clarify the additional roles of frames outside 'events'.

Other properties of eTS

Bipartitions in eTS—eTS have other peculiar properties. One of the more interesting relates to time-varying community structure in eTS. Naively, one might imagine that the number of communities should vary over time; studies using other measures of tvFC have reported such an effect [38,84]. Curiously, eTS can, at most, exhibit two cohesive communities at any one timepoint, t , where nodes are partitioned into one or the other based on the sign of their activity (Figure 4A–C) [85]. More intuitively, when we calculate the eTS, $r_{xy}(t) = z_x(t) * z_y(t)$, the value is positive when the signs of $z_x(t)$ and $z_y(t)$ are the same; when their signs differ, then the cofluctuation is negative.

This observation has implications for understanding the system-level organization of the brain (Figure 4E–G). Although there is disagreement about nomenclature and topography [36], there is consensus that at a large scale there are more than two brain systems/networks. eTS analyses show that these networks can never be expressed simultaneously. Instead, systems reflect the temporal superposition of time-varying bipartitions that themselves reflect instantaneously anti-correlated groups of nodes. In addition, removing BOLD signal amplitude and aggregating momentary snapshots of bipartition structure almost exactly reconstitutes classic correlation-based FC, and the reduction to binary signals can facilitate

computationally efficient optimization [51,86] and machine learning approaches leveraged at neural time-series [75,87,88].

Two paths to the same confluctuation matrix—Another interesting and related observation concerns the symmetry of confluctuation matrices. In general, if one were to observe only a confluctuation matrix, it would be impossible to unambiguously determine the activity pattern that gave rise to it. This is because $r_{xy}(t) = z_x(t) * z_y(t) = -z_x(t) * -z_y(t)$ (Figure 4D). In other words, flipping the signs of the regional time-series gives rise to an identical eTS. This has implications for our understanding of the relationship between FC and activity. Namely, it suggests a disconnect between the two modalities such that the sign of activity discloses little about the strength of coupling.

eTS exhibits fast autocorrelation—How quickly do confluctuation patterns change over time? This question can be addressed by calculating the autocorrelation function. Compared to sliding-window estimates of tvFC, eTS quickly decays to a baseline level, often within a few frames (Figure 4H) [47]. This matters for state-based analyses of tvFC, wherein frames are clustered into approximately recurring 'states'. One can then calculate the probability of transitioning from the current state of the brain at time t into a new state at $t + 1$. Not surprisingly, the transition matrix for sliding-window tvFC exhibits prominent diagonal structure. By contrast, brain states estimated from eTS exhibit more pronounced off-diagonal transitions – in other words, a greater likelihood of transitioning into a new state.

System-level eTS—Instead of representing all possible edges, eTS can be calculated for select subsets. For instance, in a recent paper [89] the authors extracted eTS for within-system edges [90] and compared the timing of events across brain systems. Interestingly, event timing was variable across systems and was structured along a sensorimotor–association axis. This approach facilitated the construction of 'system-triggered confluctuation patterns' – characteristic patterns of confluctuation for each system based on the timing of its events. Further, from these patterns we can also estimate edge communities by assigning each edge a community according to the system whose events elicit the greatest confluctuation magnitude.

Optimizing brain–behavior associations using eTS—Although underexplored, the perhaps most useful strategy for using eTS is to use it as a filter. Because eTS are decompositions of static FC, this means that we can selectively recombine frames to generate approximations of FC. One possibility is to stratify frames by amplitude and reconstruct FC for each level, and then compare the reconstruction with static FC. In principle, however, there are other heuristics by which we could aggregate frames and other objective functions that we could seek to optimize. One avenue is that one could reconstitute FC to maximize the magnitude of identifiability [75], brain–behavior associations, or the accuracy with which brains are assigned to a control or clinical group.

Effect of processing pipelines of eTS and events—It is well established that variation in processing pipelines can impact on FC. One of the most contentious decisions is whether or not to orthogonalize voxel/grayordinate time-series with respect to the global

gray matter signal (global signal regression, GSR). This procedure helps to reduce the impact of nuisance variables such as motion, but can also remove meaningful signal because the global signal correlates with physiological measures such as arousal [91]. What effect does performing GSR have on eTS and events? This question was partially addressed in a recent study [53]. The authors detected events irrespective of whether GSR was performed. However, the confluctuation patterns expressed during events were sensitive to this decision. With GSR, 'event states' were always composed of anticorrelated groups of brain regions – in other words, each state was made up of one set of nodes whose activity instantaneously deflected above their time-averaged mean, and another whose activity was below their mean. In the absence of GSR, these same groups appeared. However, they did not appear as anticorrelated pairs, but independently.

Linking edge-centric approaches to the broader literature

Over the past decade several studies have presented methods or reported findings that are closely related to those described in the preceding text. For instance, several of the observations made using eTS have direct precedents in the literature [92]. For instance, a decade-old study [49] reported synchronous bursts in voxel-level activity, eventually leading to the proposition that fMRI data could be described by using a point process ('active' and 'inactive' states [51]). Other studies have reported similar findings not only using fMRI data [48,49,93] but also other imaging modalities including neuronal firing [94], high-density electrophysiological recordings where they may be linked to arousal [95], and chronic intracranial recordings where they delimit behavioral state [96].

Methodologically, multiplication of temporal derivatives (MTD) [97], coactivation patterns (CAPs) [48], and leading eigenvector dynamics analysis (LEiDA) [98] are related to eTS. MTD calculates the difference in regional time-series – namely $X_{i(t+1)} - X_{i(t)}$, where $X_{i(t)}$ is the activity of region i at time t – before calculating their element-wise product. CAPs identify high-amplitude peaks in the **activity time-series** of a seed region, extract whole-brain maps during those peaks, and subsequently cluster them into a small set of 'states'. These high-amplitude activation maps can be transformed into confluctuation matrices by calculating their outer product. LEiDA is highly similar to eTS but operates on phase time-series rather than on z -scored activity, and calculates instantaneous phase differences between two sensors or regions rather than their element-wise product. In addition, none of these frameworks make explicit the link between instantaneous confluctuations and static FC. Notably, there are other instances where the eTS was described identically to that of the edge time-series framework. One recent report was overlooked due to differences in nomenclature (eTS was referred to as 'instantaneous connectivity') [99]. In another case the work was presented as a poster but never published [100].

eTS can be used to detect overlapping communities – either by applying clustering algorithms directly to the time-series themselves [101] or to the eFC matrix. Although overlapping communities are detectable using other methods [30,31,33], those methods tend to generate partitions that, although they allow for overlap, nonetheless leave most nodes with clear affiliations to a single community [102–104]. By contrast, edge communities are pervasively overlapping, such that overlap is the rule rather than the exception. That

is, most, if not all, nodes participate in multiple communities. Adjudicating between different community detection techniques is challenging and, in the absence of ground truth, generally impossible [105], further motivating the exploration of both non-overlapping and overlapping communities derived from multiple methods to probe their intersections as well as their differences.

Edge-centric approaches can also be used to identify and characterize polyfunctional network hubs. Previous studies using node-centric networks and graph-theoretic measures have identified high participation hubs – regions that connect many brain systems to one another – in association cortex and control networks [39,41]. The first edge-centric paper analyzed eFC data and clustered edges into communities [24]. An entropy measure was calculated for each brain region based on the diversity of its edge communities. In contrast to previous literature, the highest levels of entropy were concentrated in sensorimotor and attentional networks. Following up this analysis, the authors treated edge communities as a list of nodal features, and calculated a measure of feature similarity for every pair of regions. Interestingly, although sensorimotor/attentional networks exhibited the greatest levels of diversity, nodes within those systems exhibited highly similar edge community profiles. Control network nodes, on the other hand, were highly dissimilar. These observations suggest that the observed polyfunctionality of association cortex may reflect heterogeneity of edge community profiles across regions, as opposed to individual regions whose connections are distributed across distinct brain systems.

Interestingly, nodes with the highest level of overlap – those whose edges were uniformly distributed across the detected communities – included regions in the sensorimotor and attentional networks. By contrast, previous studies reported that the highest levels of overlap involved association cortex and control networks, in agreement with the view that these systems subtend a range of brain/cognitive functions [39,41]. This polyfunctionality of association cortex can be recovered from edge communities by considering the system-level variability of the edge community assignments of the nodes. Although individually nodes in sensorimotor/attentional networks exhibit high levels of overlap, their edge community profiles are homogeneous. By contrast, regions in control networks exhibit much lower overlap, but have highly dissimilar edge community profiles [24,27]. Together, these observations suggest that association cortex (control and default mode networks) may inherit its polyfunctionality not from high levels of overlap but from high levels of internal dissimilarity.

Concluding remarks and future directions

What comes next for edge-centric networks? Edge-centric networks attempt to access higher-order features of neural time-series data. Other approaches attempt to do the same [106–109].

Hypergraphs allow an arbitrary number of nodes to interact through a single 'hyperedge' and can be compactly represented by an incidence matrix, H , whose rows and columns correspond to hyperedges and nodes, respectively. For a given hyperedge (row) the associated nodes are assigned values of 1 whereas all the others are assigned values of 0.

0. These higher-order interactions can be projected onto nodes by calculating H^{TH} . The elements of this matrix correspond to the number of times pairs of nodes are linked by the same hyperedge. Interestingly, the bipartitions described previously can be reinterpreted through the lens of hypergraphs [85]. Each bipartition defines two hyperedges (one for each community), and the coassignment matrix (Figure 4F,G) is equivalent to the projection described in the preceding text.

Two alternative frameworks originate in information theory and topology. Mutual information, which quantifies the amount of shared information between pairs of variables, can be extended to include multiple variables [110,111]. Similarly, algebraic topology is a mathematical field concerned with the formation of cliques (fully connected subgraphs) and the structures formed by interacting cliques [112,113]. Both approaches are now frequently applied to neuroscientific datasets [114–118].

Although approaches for studying higher-order interactions have gained traction over the past several years, there are several limitations. One of the greatest concerns is the interpretation of eTS as 'dynamic'. This concern is not new. Previous studies have reported that stationary generative models of BOLD time-series exhibit time-varying features consistent with those of empirical recordings. This model demonstrated that mentation was not a necessary condition for tvFC, calling into question the hypothesis that tvFC reflects ongoing thought [119].

Recently, similar concerns were expressed about eTS [120–122]. One such report showed that high-amplitude events naturally manifested under a stationary model with a fixed covariance structure [120], whereas another [121] called into question whether observed network 'states' reflected statistical fluctuations. A deeper exploration of the mathematical origins of edge-centric methods showed, further, that many time-varying properties of edge-centric networks could be analytically derived (or at least approximated) from features of the static FC matrix [122]. These included not only high-amplitude events in eTS but also some properties of the eFC matrix, including edge cluster similarity. They also provided a mathematical basis for an observation made in some of the earliest edge-centric analysis [52] – namely that the global co-fluctuation amplitude derived from eTS was tightly correlated with an analogous measure made on nodal time-series, emphasizing the need for future studies to clearly establish which features of eTS are not directly anticipated from static FC or node-level properties.

These papers are essential, forcing the field to reckon with the minimal set of assumptions necessary for generating outwardly complex phenomena. In this case, the correlation structure and slow, serially correlated time-series of the brain yield realistic time-series. However, these models only demonstrate the sufficiency of those constraints [123]; their success in explaining particular properties does not imply that the brain realizes that model or that it explains all properties. Another limitation of these models is their treatment of FC as the foundation of a generative process rather than its end-product. An alternative view is that correlations do not causally shape brain activity. Instead, FC is a summary statistic that describes the history of an anatomically constrained dynamical process. That is, the structural connectivity of the brain shapes communication cascades and induces correlations

in the activity of pairs of brain regions. In this view, FC is a consequence of the dynamics rather than a causal component [124,125].

Nonetheless, a key challenge for edge-centric analyses is to demonstrate that time-varying fluctuations are not merely statistical artifacts. Several recent studies help to stave off this concern. The first showed that tvFC was predictive of spontaneous behavior above and beyond brain activity [126], suggesting a unique explanatory role for tvFC. Second, analyses of movie-watching data have found strong inter-subject tvFC [127,128]. If tvFC represented stochastic fluctuations, we would expect inter-subject correlations to be minimal. On the other hand, inter-subject correlations between edges could be spuriously driven by nodal activations [129], although regression strategies for reducing activation-based confounds in movie-watching data remain uncharted.

Owing to their increased dimensionality, the amount of memory needed to store and manipulate eTS and eFC is much greater than that of nodal time-series and static FC. To justify this orders-of-magnitude increase in computational complexity, studies using edge time-series should demonstrate that the reported effects are not easily accessible from nodal time-series alone. A clear example of this comes from a very recent study [130] in which the authors used linear models to predict spontaneous fluctuations in attention using edge time-series. Importantly, they also showed that the highly predictive edges could not easily be identified based on node-level fluctuations alone, suggesting that eTS uniquely predicts attention fluctuations.

To date, nearly every edge-centric study has analyzed human fMRI data ([64,131] for alternative approaches). Extending these tools to model organisms is an important next step. Unlike human participants where invasive experimentation presents ethical concerns [132], optogenetic and chemogenic perturbations can be used to assess the roles of individual nodes and edges directly [133,134]. Similar techniques could be used to clarify areal/population contributions to the initiation/termination of events and causally probe edge community structure.

Relatedly, future work should focus on extending edge-centric analyses to other imaging modalities. It has become possible to obtain (near) whole-brain recordings at the single-cell level in behaving model organisms [135–137], allowing direct comparison of metrics derived from eTS with ongoing behavior. In addition, the development of dual-imaging techniques makes it possible to simultaneously collect BOLD and calcium fluorescence images to compare edge-level metrics [138].

In addition, future studies should focus on applying edge-centric analyses to task-evoked recordings. Most existing studies have focused on task-free paradigms – namely resting state and movie-watching. Are there events in task fMRI studies? If so, can we disentangle which events are driven by the stimulus versus those that might occur spontaneously? How similar are cofluctuation patterns observed in task studies to those estimated from task-free designs? Does task difficulty covary with the properties of events and eFC – for example, global cofluctuation amplitude and edge community overlap?

Brain networks are multimodal and multiscale. Understanding their organization and function in health and disease is an ongoing challenge. Although existing frameworks for studying networks have facilitated fundamental discoveries, many questions remain unanswered (see Outstanding questions). Addressing them may necessitate a shift in perspective and the development of new modeling frameworks. The edge-centric model reviewed here represents one such framework. It should be viewed not as a set of tools that supersede existing methods for studying activations and node-centric networks but as a complementary approach that may offer unique insights that are not easily accessible using existing methods.

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Glossary

Activity time-series

recordings made at the level of nodes; blood oxygen level-dependent (BOLD) fMRI, the electrocorticogram, and calcium fluorescence are examples of activity time-series.

Brain network states

patterns of time-varying connectivity that recur across time and are typically detected using clustering algorithms.

Community structure

decomposition of a network into meaningful subnetworks. These are typically detected using data-driven algorithms where communities/modules are assumed to be assortative (internally dense, externally sparse).

Edge functional connectivity (eFC)

pairwise statistical interactions between eTS.

Edge time-series (eTS)

a transformation of an activity time-series such that, instead of tracking the activity of nodes over time, it tracks the cofluctuations between node pairs – namely edges.

Edge weight

the magnitude and/or sign of the connection between pairs of nodes.

Events

infrequent, brief, and high-amplitude bursts in coactivity; events are typically estimated based on the global amplitude (root mean square) of an eTS.

Functional connectivity (FC)

pairwise statistical interactions between node-level activity time-series that are typically operationalized as correlations.

Higher-order network structure

interactions involving multiple network elements; includes interactions between pairs of edges.

Line graph

a method for transforming a node-centric network into edge-centric network by linking edges to one another if they share a stub node.

Link similarity

a method for transforming a node-centric network into an edge-centric network by assigning a connection weight between edges with shared stubs based on the overlap of their unpaired stubs connected neighbors.

Structural connectivity

pairwise interactions between network nodes, such as synaptic contacts between cells, axonal projections between populations of neurons, or white-matter bundles between areas/regions.

Time-varying functional connectivity (tvFC)

estimates of changes in functional connectivity across time; typically estimated using sliding-window methods.

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Highlights

New higher-order models of brain networks can provide new insights into brain organization and function while helping to address outstanding questions in network neuroscience.

Edge-centric models shift the focus away from neural elements (nodes) onto edges.

Edge time-series are an exact decomposition of correlation-based functional networks and facilitate the detection of overlapping communities while offering estimates of time-varying connectivity at framewise temporal resolution.

Box 1. Node- versus edge-centric networks

Every network is made up of nodes and edges – the elements of a system and their pairwise interactions. However, the precise definition of these components is usually left to the researcher. This is an important decision in any network analysis because node and edge definitions shape the structure of a network and help to determine exactly what insights we might glean by studying it [139].

In neuroscience, defining nodes and edges usually means choosing whether to model connectivity at the level of voxels or parcels (fine versus coarse scale) and choosing a connection modality and an appropriate measure for assessing the presence/absence and weight of interactions (structural vs functional connections; streamline count or microstructural properties in the case of structure; correlation, coherence, phase-based measures, or information-theoretic measures in the case of function).

In addition, another dimension to consider when defining a network is the order of interactions. In most networks, edges link pairs of nodes to one another. On the other hand, higher-order networks consider edges that link groups of nodes to other groups. In general, the size of these groups is unbounded (up to the number of nodes in the network). When those groups are defined as dyads (edges), then the network model may be considered to be 'edge-centric' in that each group constitutes a network edge. Note that we recover the traditional 'node-centric' network when the groups are defined to be individual nodes.

In the same way as the choice of parcellation or connectivity metric impacts on the structure of a network, the decision to focus on higher-order interactions can yield unique insights. A growing suite of approaches are available for studying higher-order interactions in networks generally [110,113], and other approaches have been specifically developed for brain network data [24,108,109]. As more studies use these approaches, the precise contributions of higher-order networks and their role in neuroscience for understanding brain organization and function will become clearer.

Box 2. Deriving edge time-series and edge functional connectivity

How might we generate an edge-centric network that not only captures the spirit of line graphs/link similarity but is also compatible with neural time-series data and self-consistent with established methods for defining node-centric brain networks? Consider, for a moment, a single functional connection. Although there are many measures for estimating its weight, in the human neuroimaging literature, functional connectivity (FC) is usually defined as the bivariate product-moment correlation. Given the activity time-series, $X = [X(1), \dots, X(T)]$ and $Y = [Y(1), \dots, Y(T)]$ of two nodes (Figure 1D), the procedure for calculating their correlation is straightforward:

- i. z-Score each variable as: $x_z = \frac{(x - \mu_x)}{\sigma_x}$ and $y_z = \frac{(y - \mu_y)}{\sigma_y}$, where μ_x and σ_x are the sample mean and standard deviation of x .
- ii. Calculate the element-wise product: $x_z * y_z = [x_z(1) * y_z(1), \dots, x_z(T) * y_z(T)]$
- iii. Sum and normalize to obtain the correlation coefficient:

$$r_{xy} = \frac{1}{T-1} \sum_t x_z(t) * y_z(t)$$

Suppose, however, that we were to omit that final summation step. Doing so means losing the correlation coefficient and the estimate of FC. However, from (the absence of) this operation, we obtain a new construct: the cofluctuation or edge time-series $r_{xy}(t) = [x_z(1) * y_z(1), \dots, x_z(T) * y_z(T)]$. The elements in this time-series index instantaneous cofluctuations between nodes X and Y . The time t tells us when that cofluctuation occurs, the quantity $|r_{xy}(t)|$ tells us the amplitude of the cofluctuation, and the $\text{sign}(r_{xy(t)})$ tells us the valence of the cofluctuation (whether X and Y are fluctuating in the same direction or not).

Instead of focusing on a single pair of nodes (an edge), we can calculate the cofluctuation time-series for all pairs, resulting in a matrix, E , with dimensions $N_{frames} \times N_{edges}$. Using this set of time-series, we can calculate an edge-centric network analogously to how we calculate node-centric FC. That is, given the time-series $r_{XY} = [r_{XY(1)}, \dots, r_{XY(T)}]$ and $r_{uv} = [r_{uv(1)}, \dots, r_{uv(T)}]$, we can estimate their similarity as:

$$eFC_{xy,uv} = \frac{\sum_t r_{xy}(t) * r_{uv}(t)}{\sqrt{\sum_t r_{xy}(t)^2} * \sqrt{\sum_t r_{uv}(t)^2}}$$

Given its structural similarity to (node-centric) FC, we refer to this construct as 'edge functional connectivity' (eFC). The eFC matrix, however, is much larger than its node-centric counterpart and has dimensions of $N_{edges} \times N_{edges}$. Unlike link similarity and line graph matrices, eFC is fully weighted and signed, preserving this other essential feature of nodal FC. Notably, earlier studies have proposed analogous constructs [129,140–142],

although these studies used sliding-windows tvFC to estimate tvFC, and therefore exhibit reduced temporal resolution.

Outstanding questions

How can we tease apart stochastic 'noise' from truly dynamic features of edge time-series and other measures of time-varying connectivity?

What is the functional role of events? Do they represent distinct brain states or simply the high-amplitude tail of a common distribution?

Most edge-centric studies have focused on human resting-state fMRI. If the same methods were applied to non-human subjects and other recording modalities, would we observe similar phenomena?

How do events and edge communities depend on clinical status or task condition/state?

What is the cognitive or behavioral relevance of edge community affiliation, and how might this affiliation be modulated by different cognitive states or behavioral paradigms?

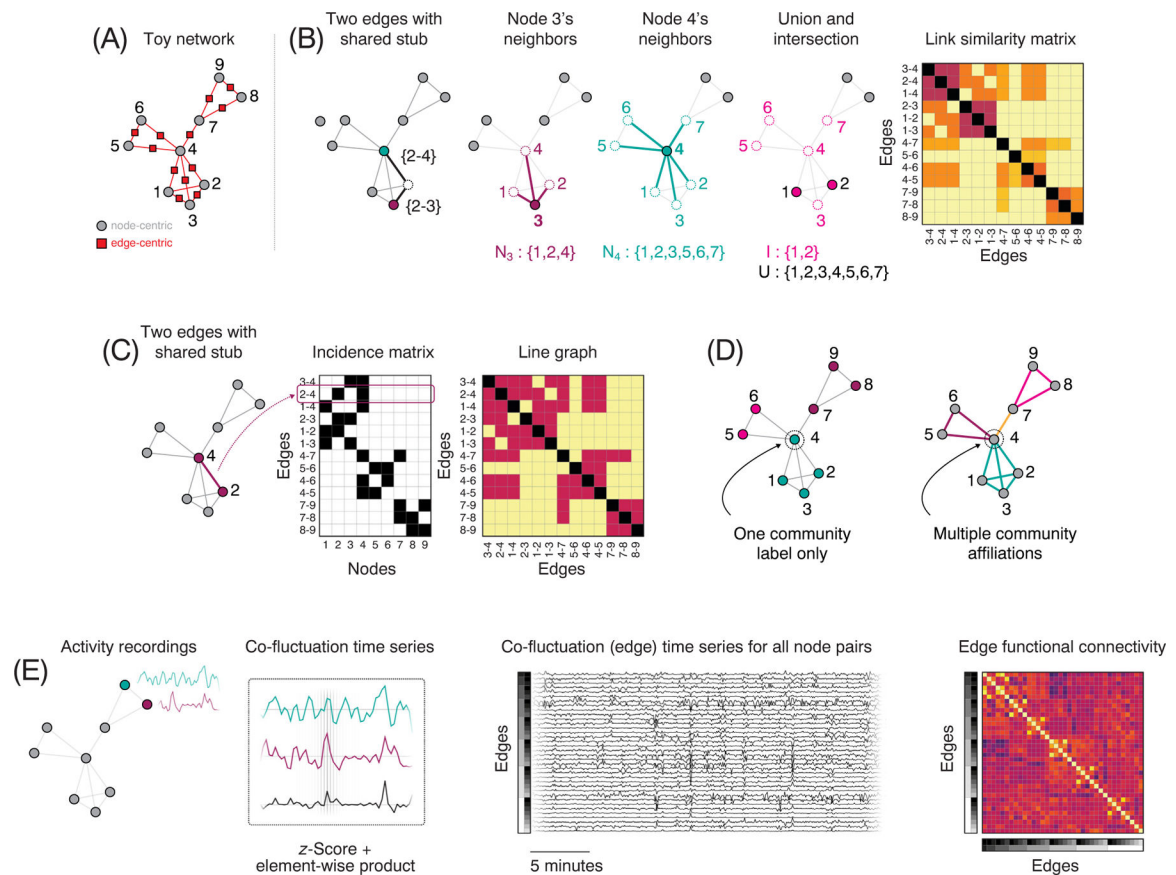


Figure 1. Transformations of node-centric networks into higher-order edge networks.

(A) Toy network of nine nodes. Node-centric and edge-centric analyses differ in that they consider interconnections among nodes (gray circles) or edges (red squares). Edge-centric analyses assign weights between pairs of edges. The 'link similarity' method considers pairs of edges with shared stubs. The weight of the connection between two edges is defined as the normalized overlap of the neighbors of the unpaired stubs. In (B) we show an example using edges {2,3} and {2,4}. These edges have node 2 in common, leaving nodes 3 and 4 as unpaired stubs. The weight of the connection between edges {2,3} and {2,4} is calculated as the Jaccard index of the neighbors of nodes 3 and 4. Another way to transform a node-level network into an edge-level network is through a line graph (panel C). This procedure involves representing the node-level connectivity as an incidence matrix. The rows and columns of an incidence matrix correspond to edges and nodes, respectively. A line graph is calculated by taking the product of the incidence matrix with itself, transposed. Edges are linked to one another if they share a stub node. (D) Community detection algorithms typically assign nodes to non-overlapping communities. When the same algorithms are applied to edge-centric networks, they assign edges to non-overlapping communities. Edge communities can be projected onto nodes, thus affiliating nodes with multiple communities. (E) Recent work has focused on analogous approaches for time-series data. Edge time-series are calculated as the product of z-scored nodal time-series. Edge connectivity measures the similarity between pairs of edge time-series.

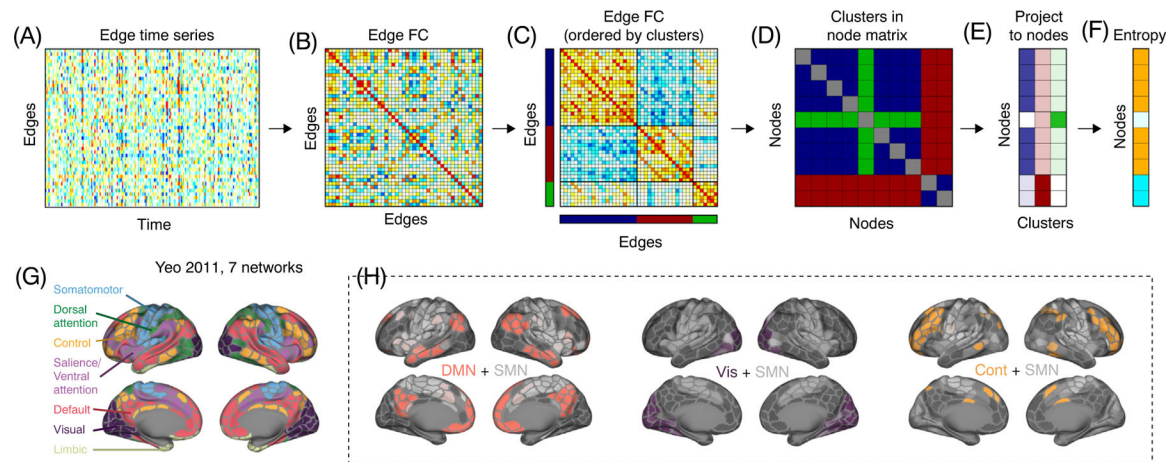


Figure 2. Edge communities and node entropy.

Edge time-series (A) are correlated with one another and can be used to calculate an edge functional connectivity (eFC) matrix (B). This matrix can be clustered using the usual matrix-clustering algorithms. Alternatively, the edge time-series can be clustered directly. In either case, we obtain cluster assignments in which each edge is affiliated with a single cluster. In panel (C) we show the eFC matrix reordered by community/cluster labels. The labels are assigned at the level of edges and can be visualized by assigning edges in a $N_{edges} \times N_{edges}$ matrix to the corresponding edge community label (D). (E) To project edge communities onto individual nodes, we calculate the fraction of node edges that are associated with each of the clusters. The result is a pseudo-continuous measure of nodal affiliation with each cluster. (F) The degree of overlap of each node can be quantified by calculating its entropy. Nodes that maintain strong affiliations to few clusters will tend to have low entropy (the blue cells at the bottom of the vector). Nodes that are affiliated equally with many clusters tend to have higher levels of entropy (the orange cells towards the top and middle of the vector). In panels (G) and (H) we contrast non-overlapping communities with overlapping communities obtained using edge-centric approaches. Panel (G) depicts 'canonical' system labels from Yeo *et al.* [8]. In panel (H) we show examples of edge communities projected onto nodes. Nodes can have partial affiliation with a given community and be affiliated with multiple communities. Abbreviations: Cont, control network; DMN, default mode network; SMN, somatomotor network; Vis, visual network. Panels (G) and (H) were adapted, with permission, from [27,90].

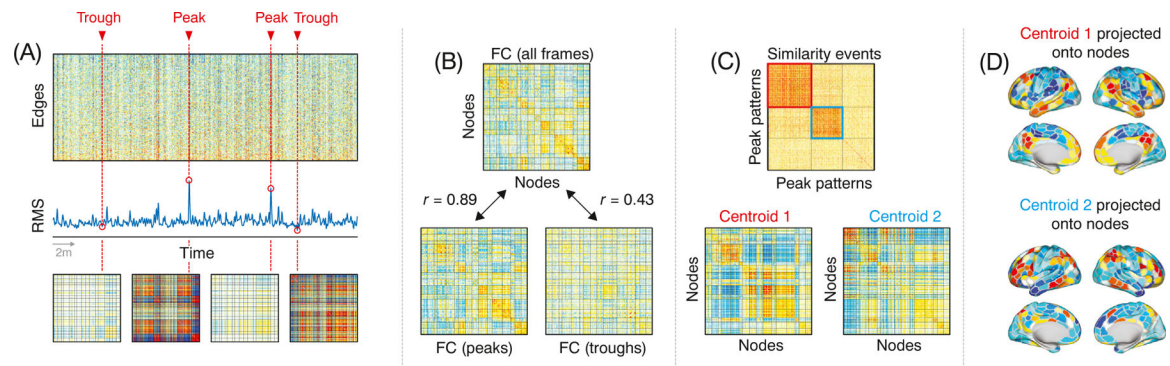


Figure 3. Decomposing functional connectivity (FC) with edge time-series.

(A) Edge time-series matrix (top). Rows correspond to edges and columns correspond to moments in time. The global amplitude of each frame is defined as the root mean square (RMS) of all cofluctuation values of all edges (middle). High-amplitude peaks in the RMS time-series correspond to 'events' and can be detected statistically by comparing the observed RMS values to those obtained under a null model in which interregional correlation structure is broken (bottom). (B) Static FC is, by definition, exactly equal to the average cofluctuation across time. It can be well approximated with a small subset of event frames; troughs yield a much poorer approximation. The approximations are generated by obtaining cofluctuation matrices from select instants in time and averaging them. (C) High-amplitude events can be partitioned into 'states' based on their spatial similarity. Cluster centroids correspond to cofluctuation matrices. (D) Matrices can be projected onto nodes to obtain 'modes' of activation that underlie the observed cofluctuation pattern. These projections are generally obtained through an eigendecomposition of the cluster centroids.

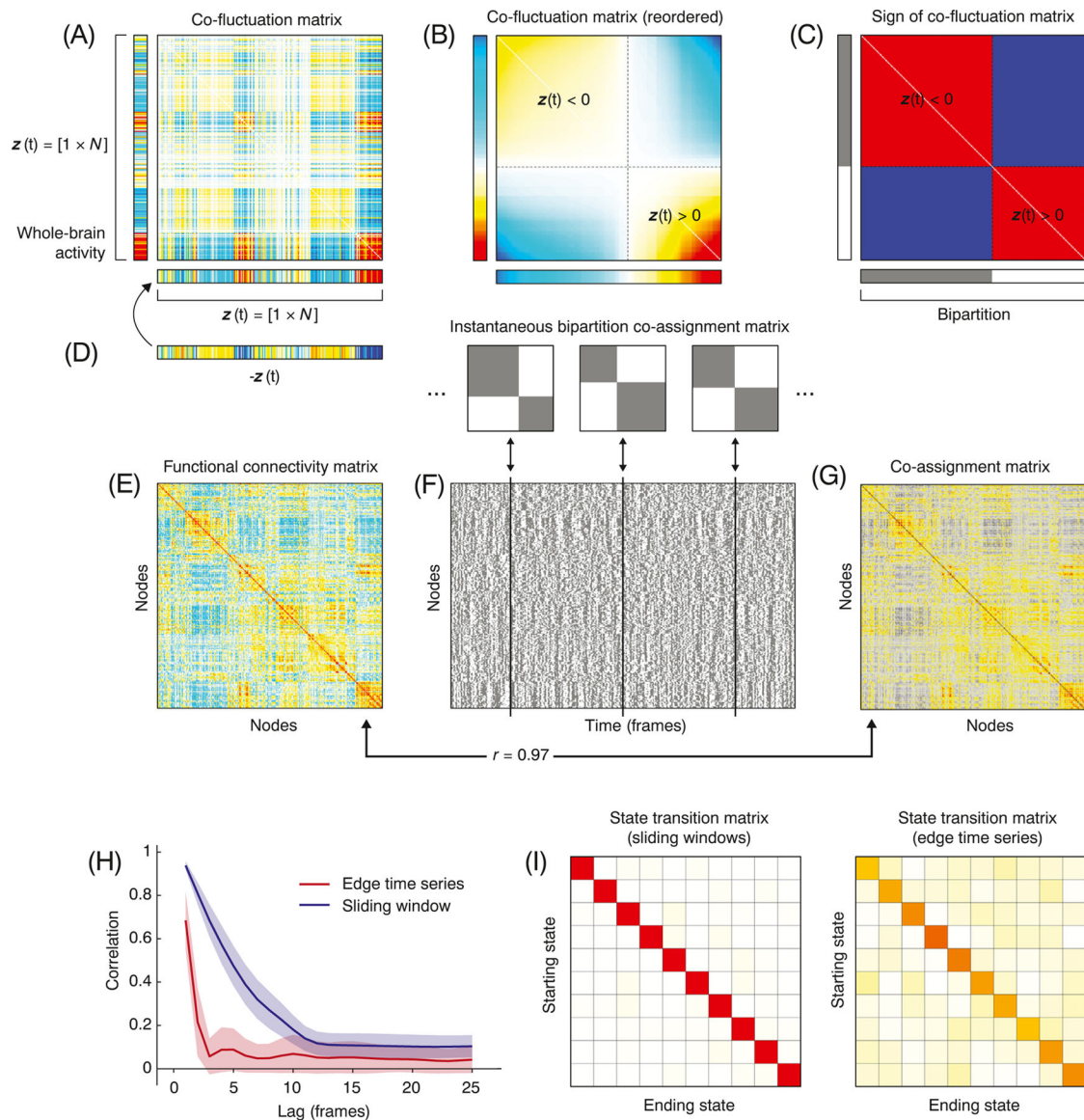


Figure 4. Other properties of edge time-series.

(A) Brain activity can be represented as a column vector of length N , where N is equal to the number of network nodes. Cofluctuation matrices are the product of a whole-brain activity vector with itself transposed. However, the same activity pattern negated (multiplied by -1) will yield an identical matrix (panel D). (B) Cofluctuation matrix reordered by the amplitude and sign of the activity vector, from negative to positive. Nodes with activity of the same sign will form a cohesive module because their product with one another is always >0 . (C) Because there are only two possible signs, there are only two possible communities, leading to a bipartition. (E) An example of a static functional connectivity (FC) matrix. (F) The bipartitions obtained from the positive/negative components of a cofluctuation matrix can be obtained more directly by taking the sign of the activity of each node at a given instant. The positive and negative elements correspond exactly to the bipartitioned components. Unsurprisingly, bipartitions change over time as activity fluctuates (F). Bipartitions can

be used to construct a time-averaged coassignment matrix. The elements of this matrix are highly correlated with static FC. (H) Edge time-series exhibit faster autocorrelation compared to sliding-window estimates of time-varying functional connectivity (tvFC). (I) This has implications for 'state detection' analyses. Network states derived from sliding-window tvFC are 'stickier' and exhibit a tendency to remain in the same state over multiple frames. In the case of edge time-series, there are more cross-state transitions.

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